Applying telepresence to education

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This paper explores the application of telepresence principles to the design and development of selected education applications. The applications are described, and the main implications for telepresence principles are explored. The work reported suggests that the overriding concentration on quality or fidelity as the most important determinant of presence may be misplaced. Indeed there may be situations where high fidelity systems may actually reduce learning effectiveness. To fully understand and model such complex situations requires conceptual frameworks to help bridge theory and practice, a varied range of technologies from which to choose, and a set of design principles or heuristics to help build the most effective systems.

1. Introduction

A Victorian time traveller visiting us today would be amazed at the use of advanced technology in today's hospitals, homes, offices and factories. However, a glance at our schools and colleges would suggest very little has changed, at least in our delivery methods. Education is under increasing pressure. It is being accused of failing to deliver quality or to embrace efficiencies. Our classrooms and lectures halls are becoming overcrowded, the numbers are increasing but resources are remaining stable or indeed in some sectors reducing. Against this rather depressing backcloth many are seeing technology as the new saviour [1, 2]. However, too often, technology is advocated solely for increasing efficiency, whereas, it is argued here that its real power lies not just in efficiency savings but also in improving the quality of education.

The concept of presence and therefore telepresence is not simply an issue of 'being there'. There are circumstances, particularly in education, where the learner is present physically but is mentally miles away! Psychological and physical distance are not the same thing — many campus-based students will have little or no sense of 'presence'. For learning situations we need to stretch the concept of presence to include 'psychological' presence which could cover such concepts as 'engagement', 'relationships' and 'audience'. For learning, it is critical for the individual to engage in the learning process — this ensures motivation and assumes that the learner is grounded in the concepts being taught. If a learner cannot relate to the new concepts then they will always remain distant and probably misunderstood. Also, presence is often associated with active participation — being there is not sufficient, you also have to contribute. Yet it is well known that 'mere presence' or the presence of a passive audience can motivate people, and simply knowing others are there may be as important or even more important than the individual being there.

Relationships are important because of their role in determining expectations during conversations or communications. For example they help determine the interaction patterns or communication structures. In Carey's five models of information flow [3], the structure of the interaction pattern is determined by the different roles and relationships of the participants (e.g. in the 'classroom' model, the moderator role assumes authority that is manifested in a controlled one-to-many interaction pattern). Relationship can also provide insight into the purpose of the interaction. People rarely communicate without some purpose or topic of communication. This does not rule out the casual or informal meetings which often have the purpose to explore or update understanding [4]. Relationship may also suggest the form and content of the interaction. Artefacts, either conceptual (e.g. an idea) or real (e.g. an object), are important for supporting or 'scaffolding' the conversation. The ability to easily access and share such artefacts could be critical in learning situations, e.g. through the use of 'whiteboards', document cameras, etc. The nature of the relationship may also influence the formality or emotional tone of the communication, e.g. mother to child or boss to employee relationship. More critically relationships also determine the nature of task interactions and the subsequent dialogues around those tasks, e.g. learner/teacher, designer/implementor relationships.

This paper will present a vision of education based on sound principles of learning and educational psychology. A vision that essentially sees learning and teaching as a social and collaborative process, and a process ripe for the application of telepresence principles. The focus will be on educational technology in general and not just to support distance learning. The paper covers work being carried at BT Laboratories to test these ideas by building robust prototypes and by using them in real educational settings, e.g. schools and colleges, and will also provide a glimpse of the future where we will be less inhibited by bandwidth and processing constraints. Finally, there is a discussion on what has been learnt from these examples and how they have changed people's perceptions of the importance of telepresence to the learning context.

2. Understanding education

There are, at least, two competing paradigms or approaches found in the education literature [5]. One approach, the didactic or instructionist approach, advocates that the teachers possess the knowledge and understanding and have the difficult job of 'transferring' that knowledge from their heads into the heads of their students. In contrast, a constructionist approach argues that effective learning is best achieved when the student is in control of their own learning. Figure 1 summarises the main differences between the two paradigms.

However, a new paradigm is not sufficient. There is a need for a way of linking our understanding of learning with appropriate technologies. A project was commissioned to create a conceptual framework to help us bridge the pedagogy with the technology [7]. The framework assumes that :

- knowledge is situated — it is situated in action, and the meaning of the action is itself determined by task and social environments; we use knowledge, and without that use it has no or limited value,
- knowledge is relative — understanding is relative to what you already know,
- knowledge acquisition is a dynamic and iterative process — it involves a conceptualisation cycle where new ideas are continually tested and refined, and the refinement process changes our new knowledge state, thus producing a reconceptualisation,
- knowledge can be viewed as a set of interrelated concepts.

From these basic assumptions three top-level learning stages are offered:

- conceptualisation — the coming into contact with other people's concepts,
- construction — the building and testing of one's knowledge through the performance of meaningful tasks,
- dialogue — the debate and discussion that results in the creation of new concepts.

It is important to note that conceptualisation is about 'other people's concepts', that construction is about 'building knowledge' from combining your own and other people's concepts into something meaningful, and that dialogue refers back to creation of 'new concepts' (rather than knowledge) which then triggers another cycle of the

	instruction	→	**construction**
classroom activity	teacher-centred didactic	→	learner-centred interactive
teacher's role	knowledge provider always expert	→	knowledge receiver sometimes learner
learner's role	listener always learner	→	collaborator sometimes expert
educational emphasis	facts memorisation	→	understanding discovery
concept of knowledge	accumulation of facts	→	transformation of facts
success criteria	quantity retained	→	quality of comprehension
assessment	norm-referenced written examinations	→	criterion-referenced project portfolios
use of technology	drill and practice	→	communication, collaboration, presentation information access and

Fig 1 Paradigm shifts: from instruction to construction (adapted from Dwyer [6]).

conceptualisation process. However, the framework does not just assume that dialogue only takes place at the dialogue stage. There will be dialogue for clarification at the conceptualisation stage, and dialogue of collaboration at the construction stage.

3. Applying telepresence in the school sector

In some earlier studies with primary school children using videoconferencing technology (the PC-based VC8000 over ISDN-2) it became clear that the constructionist approach is extremely effective in increasing children's levels of motivation, their general attitude to learning, and their levels of self-esteem. It also became clear that within this particular context the videotelephony element had a powerful social role to play in the children's interaction (the 'hi' and the 'bye'), but added little to the successful completion of their particular tasks. This not an uncommon finding [8], and increasing the level of presence is unlikely to change the finding. For the particular collaborative tasks undertaken by these children it was the dataconferencing facilities (e.g. whiteboards, application sharing, and the use of file transfer protocol (FTP)) in conjunction with high-quality audiotelephony that were critical to their efficient and effective completion of their tasks.

The children also became experts in specialist areas (e.g. in the use of spreadsheets) and soon were 'teaching' other children and their teachers. For these children to be seen and valued as experts was an important contribution to the high levels of self-esteem and subsequent high motivation levels. It was also found that children liked the fact that there was a wider audience, beyond the teacher and their parents, who could see their work. Again this sense of audience was a powerful motivator to produce quality materials of which the children could feel justly proud and wish to share. The effects of audiences on performance has a long psychology pedigree and can be traced back to the pioneering work of Allport [9]. A fuller description of the videoconferencing work can be found in Fowler et al [5].

In the subsequent years it was decided to increase the number of schools to include some secondary schools and to explore the application of these concepts to an Internet-based service. In line with the constructionist approach, children were asked to spend more time authoring the Web rather than surfing the Web. To help achieve this a Netscape-based framework was set up, a range of authoring tools (e.g. AOL press, MacroMedia Director), and some data collection tools (e.g. DAT recorders and Digital cameras) were provided, together with Web-based support in terms of 'hints and tips' for developing your own Web site as well as an example of good practice that the children and teachers could follow.

The constructionist approach emphasises the importance of process as well as the products of learning. To reflect this, the Web site was divided into two areas for the content. One area was for the completed products. The children placed their material in this part of the site when they felt happy enough to share it, and thus encourage other schools to enter into a collaborative partnership with them. One school, for example, has set up an international rivers project with schools participating from all over the world. The idea is for the schools to compare and contrast their local rivers. The other area is for work in progress. This area also will be public and other schools will be encouraged to make constructive comments at this formative stage of the creation process.

The level of communication, and indeed by implication the degree of 'presence' required, was thought to be best provided through a standard groupware product (FirstClass) which provides e-mail, text conferencing and 'discussion groups' facilities. These were chosen because in this particular context the children are being encouraged to be more reflective and thoughtful, and an asynchronous solution seemed best to support this level and type of communication.

One problem encountered fairly early on was the difficulty some teachers had in finding a focus that had curriculum relevance and that could be meaningfully shared across schools both in the UK and abroad. To help stimulate the project, the team in conjunction with the local newspaper (the Ipswich Evening Star) suggested a common theme of '1000 years on the Orwell'. The Orwell is a local river in Ipswich near which many of the trial schools were located. Schools were encouraged to create projects addressing all areas of the curriculum (for example, pollution on the Orwell, a collaborative novel — 'Murder on the Orwell', the history of Landguard Fort at the mouth of the Orwell, etc).

The richness of the context, and the fact that the children 'owned' and were ultimately responsible for their own work, changes the nature of any on-line or off-line communications. The off-line communication is rich in the dialogue of collaboration and negotiation. The children work in small teams and have to plan as well as do the work. The same richness and sense of ownership attenuated the need for a high level of presence for the on-line communications, making fairly low fidelity communications systems with their associated low levels of 'presence' perfectly acceptable. This was possible because of the commonality and shared understanding of the task context. Rivers do, after all, exist nearly everywhere in the world, and children are children regardless of their background, colour or creed.

4. Applying telepresence in further and higher education

When the role of technology in supporting distance learning in further or higher education (FE an d HE respectively) came to be considered, it became clear that the social aspects were not only important for supporting collaboration, but also because people like a sense of group identity (i.e. the student cohort).

Putting people, who are essentially strangers, into contact with each other is always going to be difficult. Conversation usually occurs for a purpose, and often to be successful the communicators must have a shared understanding of their subject matter. Indeed, Vygotsky [10] would argue that lovers, as an extreme case, are so close that communication between them becomes sparse and telegraphic as they already know what each other is thinking. A system was therefore needed that allowed students to quickly provide and share information on each other that could then be used to 'seed' further conversation, until at least the course itself provided enough common experiences for the students to share, and the cohort to form.

A further factor to be considered was the importance of real-time interactions in effective communications. The question yet again came down to quality or fidelity of the communications medium. The ability to appropriately sense and react to non-verbal cues is clearly important for mediating and controlling conversations. In particular the use of eye contact to control turn taking is well understood [11]. In contrast Bales [12] stresses the importance of the social and emotional activities that surround effective task-based group interactions. These can be either positive (e.g. shows solidarity, shows tension release) or negative (e.g. disagrees, shows tension), but from our perspective what is important about Bales' categories is that they are communicated equally effectively by spoken rather than visual cues.

To support these social needs, an Internet-based system for teaching English as a Foreign Language (RISE-Merlin) was developed and tested with Hull University. The screen shot in Fig 2 shows the main screen. The important features to note from a telepresence perspective is the 'People' and 'Meeting Place' facilities.

Fig 2 The Merlin notice board.

Activating the people link allowed students to access brief biographical information about their tutors and other members of the course. It was hoped that this would provide some information to help 'break the ice' in subsequent real-time interactions. These real-time interactions take place in the Meeting Place (see Fig 3).

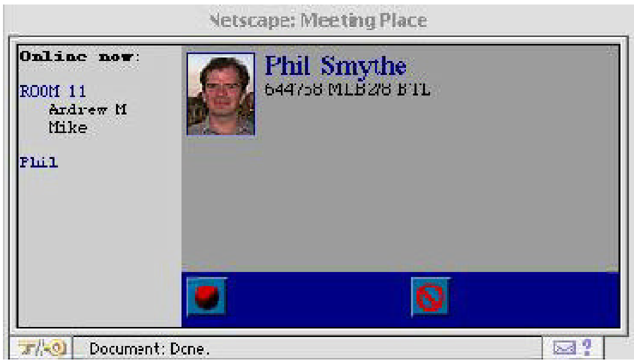

Fig 3 Merlin's Meeting Place.

The 'meeting place' allows the setting up for multi-point audioconferences from the Web browser. All students who are logged on to the system are listed under 'on-line now'. If an audioconference is already taking place the participants are placed in a 'room'. Others can be invited to join the conference or even denied entry into the room. There was a need to ensure the use of high-quality speech both for teaching purposes and for maintaining acceptable feelings of presence. It was felt that the current versions of the Internet phone could not consistently deliver the required quality, and so the PSTN facilities were used. This does have the implication that students and teachers need two 'lines' — one for data and one for voice. This could be implemented by two PSTN lines one for the telephone and the other for a modem (minimum 33 kbit/s), or an ISDN-2, or LAN access and a telephone.

One of the outcomes of the first trial was that people were reluctant to engage in chance meetings. They preferred to know in advance who was on-line and when. In the second version of RISE-Merlin a diary was added (see Fig 4) so that people could schedule their meetings.

Fig 4 The diary.

RISE-Merlin also provides more traditional learning support (see Fig 2). The ‘pathways’ contain most of the teaching content provided by Hull University. Hull also provided materials for ‘resources’ which could be material specific to English as a Foreign Language (e.g. dictionaries) or interesting links to other Web sites which could encourage conversation and debate. Finally RISE-Merlin’s ‘portfolio’ provides facilities for students to store their exercises either as personal documents or as shared (group) documents. The ‘documents’ can be a mix of text or audio files. The audio files could be, for example, a tutorial recorded in the meeting place.

The two technically unique features of the RISE-Merlin prototype were the use of a dynamic webserver which allowed access to data specific to a student and a course. To ensure this, the students have to log on and provide a password. The system can then access materials relevant for their course and files personal to them. The second unique feature is the ability to set up an audioconference from a Web page using high-quality PSTN services rather than relying on the more variable quality of the Internet phone.

RISE-Merlin was tested on 35 students from Western Europe, Central Europe, the Far East, USA and South America. Of these students who undertook the 15-week intermediary EFL course, ten accessed also Merlin from their homes, sixteen at their Colleges, and six from their work. Other students had to make special arrangements to access an appropriate machine. Early results from the evaluation study suggest that the twelve triallists who completed the studies enjoyed doing the course, were satisfied with it, had received considerable help with their English, and would definitely want to do another Merlin course. The findings suggested that the students who dropped out either suffered unacceptable technical problems or tended to favour more structured classroom-based approaches, found it difficult to talk to other students on-line, and felt the need for more conventional tutor contact time. Overall, and with certain caveats, the findings suggest that for certain kinds of learners and learning situations a carefully designed and interactive on-line course can be as effective as undertaking a traditional course.

5. Beyond RISE — advanced learning environments

The work described above in schools and colleges uses today’s technology (predominantly PCs ov er ISDN) and was constrained by that technology. The conceptual framework, by definition, is technology independent, so the question was asked what an advanced learning environment (ALE) would look like if there were no bandwidth, processing or hardware constraints. In such an ideal world a high-quality or fidelity telepresence solution would become available.

The questions then are: When do we need such fidelity and how does it help us learn? The project is new and to date there are only visualisations of what a high bandwidth (100 Mb) desktop version of RISE might look like.

At the conceptualisation stage in Mayes’ framework, we imagined that small groups of learners (about three) would like to share the ‘lecture’ experience.

In Fig 5 a mock up of what the user interface may look like for the conceptualistion stage can be seen. There is a central space or ‘learning canvas’, and below that is a ‘film strip’ from which a lecture can be chosen. The boxes to the left are communication windows, and in this version they would be using high-quality multipoint videoconferencing.

Fig 5 The user interface (UI) for ALE’s conceptualisation stage.

In Fig 6, the lecture is in progress and is being watched by three students. The students are able (with mutual agreement) to fast forward, pause, rewind, etc, the lecture which is ‘streamed’ from a large central video server. They are also able to make notes while the lecture is in progress and to attach the notes to the relevant part of the lecture material. Notes can be private or shared.

If at any time a student wishes to ask a question for clarification purposes, they can stop the lecture, type in and submit their question, and a clip from a video server of the lecturer answering the question then appears (see Fig 7). If in the unlikely event no suitable answer is available then the lecturer will suggest emailing a tutor or expert for an answer. The answer will, at a later date, be added to the video database. Although technically it is possible (but difficult) for the student to speak the question, the use of a written question seemed more appropriate as it supports a more reflective and thoughtful approach. This is similar to the traditional situation where learners will often spend time composing their question either in their minds or on paper to ask the lecturer at the end of a session.

Fig 6 Sharing the lecture.

Fig 7 Asking questions at the conceptualisation stage.

Although learning in a group is important, it is most probable that ‘protocols’ for social learning will have to be developed. For example, a simple protocol may be that no student should interrupt on the first viewing of the lecture material. Only on subsequent viewing would the students be encouraged to question each other’s understanding and to use the VCR controls to access relevant parts of the lecture to illustrate or support any point being made.

At the construction stage, the learning canvas now supports a highly interactive, modelling or simulated environment. This stage could involve more traditional data-conferencing facilities to support collaborative writing or creation of multimedia materials. It could also support virtual worlds where the learners will be able to touch and manipulate objects and have the means of sharing their experiences in a collaborative way. The collaboration can be achieved in two ways — through the conventional videoconferencing windows for the small group, and in virtual worlds where others are invited from outside that group who can have virtual representations through the use of avatars.

Finally for the dialogue stage we return to traditional videoconferencing set up. Initially the tutor appears on the learning canvas and assumes the role of chair person (see Fig 8), but if a tutee wishes to take control this can be signalled and if the chair person permits they can move into the central canvas and take control of the dialogue.

Fig 8 The UI for ALE’s dialogue stage.

Notes can also be taken and shared. The video-conference can be recorded and reused either for revision purposes, or as source material for subsequent lectures.

It is also the intention to specify and build a microlearning centre. The design of such a centre will still be based on the basic principles described in Mayes’ Learning Framework, but will try to escape the constraints of the traditional learning ‘furniture’. In current distance learning technology, that furniture is represented by the desk-top computer. The microlearning centre will build a fully interactive virtual environment (IVE). The technical challenges will be to understand the interfaces between many different types of technologies from a simple telephone, through to a desktop computer, to fully emersive virtual worlds involving high-definition wrap-round projections. The educational challenges will be to design such centres to be educationally valid and to prove them to be so. To achieve this, the same content currently used for one of the BT MSc modules will be used. The content will then be converted and presented in the two new technical learning environments — the desktop and the microlearning centre. However, the effectiveness will then be measured across the three environments — the traditional, the desktop and the mircolearning centre.

6. Discussion

Our original thoughts on the relationship between presence and learning were driven by such concepts as engagement, relationships and audience. These undoubtedly proved important. On reflection the key and superordinate concept seems to be the notion of relationships. The relationship concept embodies notions of task and purpose, and also the type of relationship sets up expectations according to the roles of the participants. The nature of the relationship could also determine whether or not it is appropriate to include a wider audience as part of the communication and the degree of formality of the interaction. In formal situations a full range of cues are required and a shared understanding cannot be assumed between the communicators. In such a situation, high-fidelity systems would be advantageous. In less formal situations, less cues are required and lower fidelity systems would be acceptable.

The importance of the learning task cannot be underestimated in designing effective systems. There appear to be two important concepts. Firstly, there is the main learning 'content' which in ALE and RISE systems takes place in the learning canvas. In terms of Mayes' conceptual framework, the content of the learning both in form and type will vary according to the learner's stage in the framework (i.e. conceptualisation, construction and dialogue). The other concept appears to address the dialogue or conversations that 'go around' the learning content. In a similar fashion to content, the nature of the dialogue (i.e. clarification and collaboration) will vary according to Mayes' first two stages.

Clearly the two variables are related or bound together by the learning framework. However, in terms of applying telepresence there seems to be important differences. When dealing with the conceptualisation stage the critical issues appear to be fidelity or realism of the material. The conceptualisation stage needs to be rich and flexible, allowing learners with different backgrounds and abilities to quickly grasp new concepts. The issue is often to build upon what the learners already know and to use concrete illustrations to push the learner into the 'conceptual' unknown. Often at this stage the narrative or story line needs to be explicit and in the control of the teacher. This, in association with relative passivity required of the learner, it is argued, demands high-quality well-structured, balanced and engaging material. The necessity for engaging content suggests parallels with a well-made TV documentary, and indeed traditional producers of broadcast material may have an important role to play in educating teachers in the production of engaging learning materials.

At the construction stage the emphasis is more on 'doing'. Photorealism can be sacrificed for higher levels of interactivity with objects represented in the 'constructive' world. Realism is still required but the emphasis is now on the quality of the behaviour of the object rather than the quality of its representation. In other words, the learners should be able to interact with objects in their virtual world and the behaviour of those objects should be both internally and ecologically valid. Internal validity refers to the appropriateness of the behaviour within the virtual world, whereas ecological validity refers to the correlation between the behaviour of virtual objects with their real world counterparts (where they exist). Furthermore, in the construction stage the control of the narrative needs to move from the teachers to the learners. The role of the teacher is one of facilitation or guidance in ensuring that the narrative is made explicit in order that the learning lessons can be abstracted and generalised to new situations.

At the dialogue stage there is a movement from the concrete to the abstract. Such a movement assumes a change in knowledge state from the novice to the more expert. Dialogue between experts is often carried out in real time, has little predetermined structure (i.e. the arguments 'flow' from point to point), and often requires access to shared whiteboards and notepads to graphically illustrate points or arguments. The essential element of discourse in such situations demands flexible or dynamic communications facilities. For example, to regulate and support a highly dialectic situation requires a high-fidelity videotelephony where participants can easily sense critical non-verbal cues that will regulate the argument and indicate the emotional tone of the discussion. Equally there will be situations when the discussants will want to access whiteboards or show artefacts in order to support their positions. In such situations high-quality audio combined with data-conferencing is required.

The dialogue aspects of the framework also make differential demands on telepresence principles. Dialogue for clarification only requires a real-time presence of the questioner rather than the respondent. This is based on the principle of 'most frequently asked questions' which assumes in the vast majority of cases the questions asked are not original and a database of answers can be created and accessed. Moreover, the formulation of the question often requires reflection and so the use of text, albeit in real time, with the ability to edit before submitting, would meet the 'reflection' requirement. The reflection requirement also demands a greater temporal separation between the content that provoked the question and the question itself. It is often only after reflection that a failure to understand a concept becomes apparent. Indeed the design of the conceptualisation stage needs to support such dissociation through the ability to replay, edit and/or tag 'conceptual' material.

Dialogue for collaboration would also need to be in real time in order to map on to the current 'constructive' process, and would be symmetrical in requiring the real-time presence of all participants. The need to map in real time between the conversation and the content requires the ability for participants to easily identify (e.g. pointing), manipulate (e.g. gripping) or observe realistic changes in behaviour (i.e. good speed or latency of response is required). These behavioural elements combined with real-time verbal abilities (either voice or text) are critical to this stage. There is no requirement for the ability to record or play back collaborations other than to support clarification dialogues (e.g. 'why did you do that?') or as input (i.e. as content) into conceptualisation stage.

The marriage of the conversations with the learning stage is also important in suggesting future research directions. It helps direct our research effort away from the syntax of conversations (i.e. the rules that govern the utterances) which are becoming well understood [13], at least in video mediated communications, to the semantics of the conversations. Meaning or semantics can be best understood within the situation or context of the utterance which itself can be captured through the concept of relationships.

7. Conclusions

In the learning context at least, the continual push for higher and higher quality or more and more realistic telepresence systems is not always necessary, and indeed in some situations may distract rather than benefit the learner. The notable exception to this may be in cases where a learning task requires a high-quality and realistic visual content to be valid at the construction stage (e.g. in teaching surgery, or engineering), or again at the construction stage where in the 'doing' it is important to detect emotional changes in peoples' interactions (e.g. in teaching counselling or interviewing skills).

A more general conclusion arising from this work, is that basic telepresence principles like 'presence' or 'engagement' are best understood within applied contexts. Indeed it may be more useful for designers to have a set of heuristics (or rules) rather than principles which can then inform design decisions. More research needs to be undertaken to understand the rules governing the different types and quality of representations and communication required in different learning situations. Only then will we be in a better position to truly answer the question of whether or not 'being there' is important.

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